Cape Peninsula University Of Technology | 石华智能 (Shí Huá SmartTech) | AgriSmartSA – AI-Powered Automated Watering System

HUAWEI ICT COMPETITION 2025 – 2026 INNOVATION COMPETITION

REAL-WORLD PROBLEM

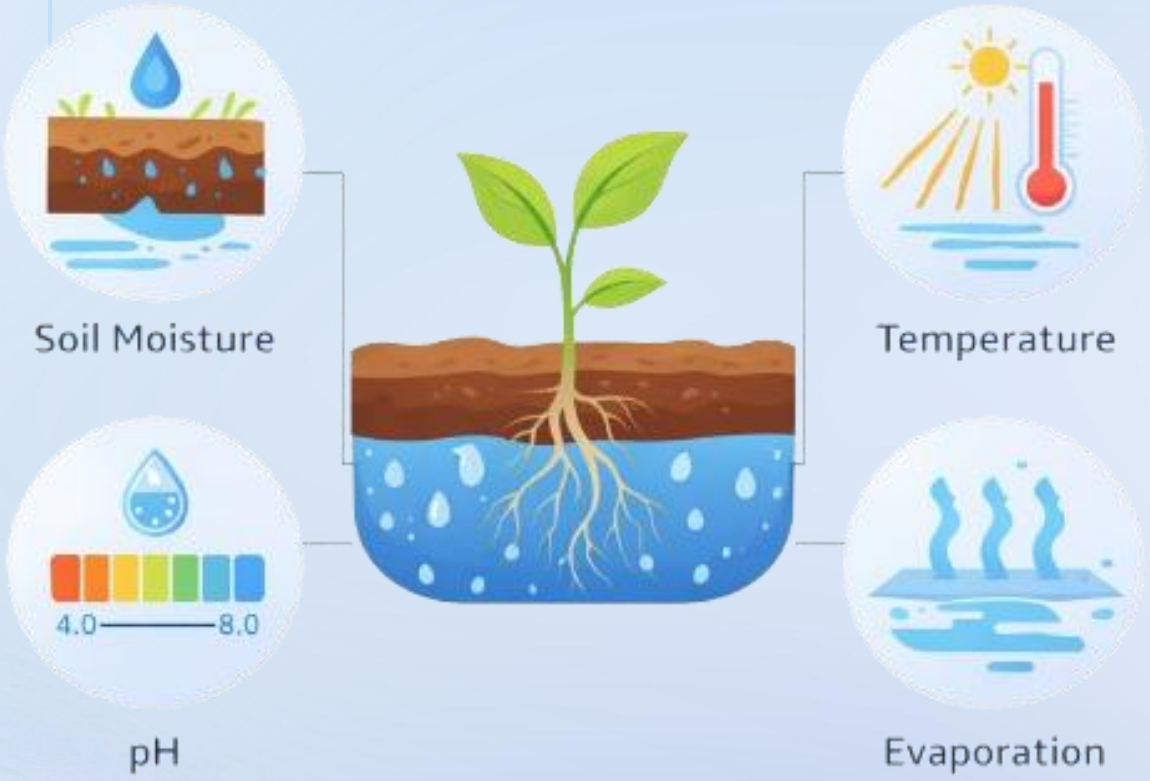
- Water scarcity & unpredictable rainfall – difficult to farm
- Manual irrigation → water waste
- Small farmers lose yield & income

Challenges :
low Crops
High water usage
High labor Cost
Limited access to technology - Without it, plants die

Farmers have no guidance on which crops need more water, less sunlight, or better care.

Farming is stressful, expensive, and risky

Why Over or Under watering happens?



Do not have the real–time information on what the plant and soil actually need.

Reasons

- Soil moisture – water availability.
- pH level – nutrient + water absorption.
- Temperature – transpiration rate.
- Evaporation – water loss.

Framers use manual irrigation and do not have the right knowledge of what the plant needs

Soil PH (Potential of Hydrogen) and temperature affect water uptake

SOLUTION = AGRI SMART SA

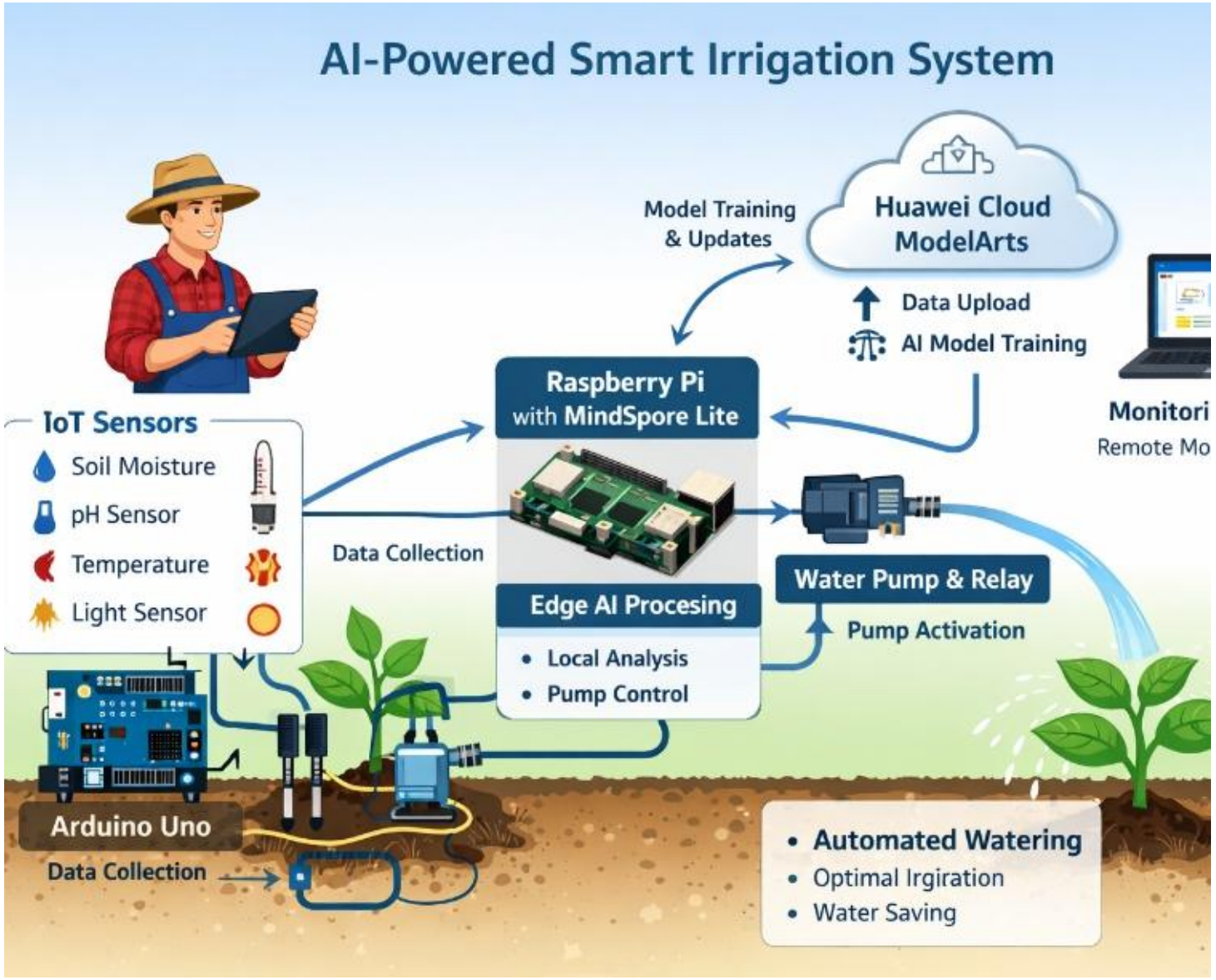
- Built with one purpose: " Help Framers grow more food while using less water. "
- No more guesswork, AgriSmart Sa Listens to the land.By monitoring real farming conditions using sensors

  Sensors
  - measure soil moisture, temperature, and humidity
  -tracks upcoming weather.
- Collected data will be stored in the app to track the readings of the plant in real time.
- Integrated AI to help the system decide the right time to water and care for crops

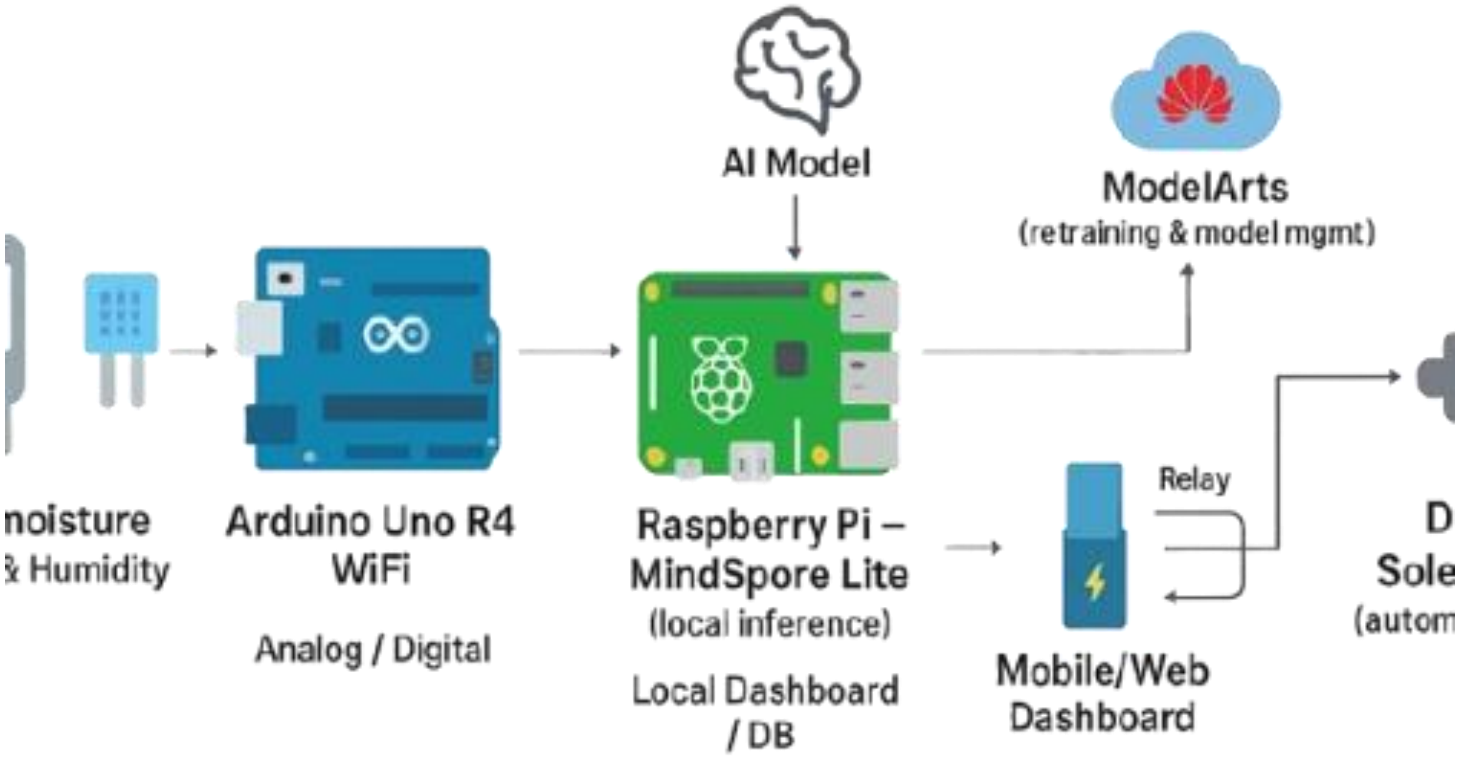
- The Impact = 40% of water saved, more healthy plants being produced, and save money, reducing labor costs.
- AgriSmart turns farming from guesswork into making smart decisions based on collected data

SOLUTION OVERVIEW

AI decides when and how much to water—automatically.



TECHNICAL APPROACH



- Offline mode: local inference + manual override
- Cloud sync for regional model improvements
- Safety: max runtime & moisture threshold hard-coded to prevent over

DATASET AND AI

DATASET

- Soil Moisture.
- Soil pH.
- Irrigation Events.

ARTIFICIAL INTELLIGENCE (AI)

- Trained on real farm data.
- Predicts watering time & duration.
- Runs Locally offline.

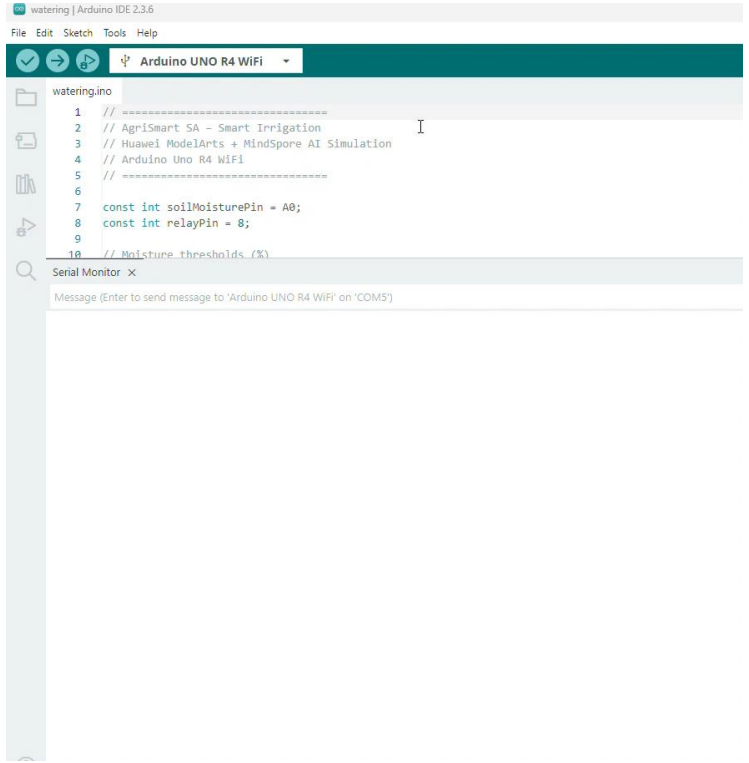
Key Code

Inputs: moisture, pH, humidity, temperature

```
void loop() {

  // Read moisture
  int rawMoisture = analogRead(soilMoisturePin);
  int moisturePercent = map(rawMoisture, 1023, 300, 0, 100);
  moisturePercent = constrain(moisturePercent, 0, 100);

  // Simulated pH
  float phValue = simulatedPhValues[phIndex];
```

Output: irrigation duration & timing

```
  // Simulated pH
  float phValue = simulatedPhValues[phIndex];
  phIndex = (phIndex + 1) % 4;

  // Get AI decision & confidence
  float confidence = 0.0;
  String decision = aiDecision(moisturePercent, phValue, confidence);

  // Display readings
  Serial.print("Soil moisture: ");
  Serial.print(moisturePercent);
  Serial.print("% | Soil acidity (pH): ");
  Serial.println(phValue, 1);

  // Display AI Module output
  Serial.println("[Huawei ModelArts AI Module]");
  Serial.print("- AI Decision: ");
  Serial.println(decision);
  Serial.print("- Confidence: ");
  Serial.print(confidence * 100, 0);
  Serial.println("%");

  // Apply pump state based on AI decision
  if (decision.startsWith("Watering recommended")) {
    pumpState = true;
  } else {
    pumpState = false;
  }
  digitalWrite(relayPin, pumpState ? LOW : HIGH);

  Serial.println("-----------------------------------------------");
  delay(10000);
}
```



DEMO VIDEO SLIDE

Scalability & Partnerships

Designed for village → cooperative → national scale.

SCALABLE DESIGN

- Modular low-cost units.
- Works 1 plot - 20 hectares.
- AI improves with more farms.

PILOT TESTING

- 10 – 20 devices.
- Subsistence & commercial farms
- Real performance data

PARTNERSHIPS

- AgriSA.
- Potatoes SA.
- Grain SA.
- DALRRD.

THANK YOU

AgriSmart SA uses Huawei AI to secure food, water, and livelihoods.